

Auto Ticket Classification using Flow Designer

Project Title: Auto Ticket Classification using Flow Designer	Course/Module: ServiceNow System Administrator
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1. Project Overview & Objective

The school IT helpdesk receives a high volume of daily support requests from students and faculty covering network connectivity, hardware failures, account access issues, and system performance bottlenecks. Currently, IT support staff manually review every incoming ticket to identify the issue type, assign appropriate categories and subcategories, and notify callers. This manual approach is time-consuming, prone to human error, and lacks scalability. This project automates ticket classification and notification dispatch within ServiceNow using Flow Designer by analyzing keywords in incoming incident descriptions.

2. Milestone 1: Requirement Analysis & Planning

Activity 1: Business Use Case

- Automatic classification of incoming tickets based on issue description.
- Dynamic assignment of Category and Subcategory without manual staff intervention.
- Dependent choice filtering between Category and Subcategory fields.
- Automated email notifications sent immediately upon ticket submission.

Activity 2: Creation of New Update Set

- Created a dedicated local Update Set named *Project Update Set* in Global application with state set to **In progress**, and set as current to isolate and track all configurations.

Activity 3: Navigation Steps

- Documented precise navigational paths across System Update Sets, Flow Designer, and Table Designer.

3. Milestone 2: Backend Development & Configuration

Activity 1 & 2: Custom Table & Field Configuration

Created a custom table with the following schema:

- **Number:** Auto Number | **Caller:** Reference (sys_user)
- **Category:** Choice (Network, Hardware, Access, Performance)
- **Subcategory:** Choice (Wi-Fi, Projector, Forgot Password, Slow Computer)
- **Short Description & Description:** String | **State:** Choice (New, In progress, On hold, Resolved, Closed)

- **Assigned Group:** Reference (sys_user_group) | **Assigned to:** Reference (sys_user)

Activity 3: Dependent Choice Logic

- Configured dictionary attributes on **Subcategory** to enable *Use dependent field* pointing to **Category**.
- Mapped subcategory choices: Wi-Fi → Network, Projector → Hardware, Forgot password → Access, Slow computer → Performance.

4. Milestone 3: Automation using Flow Designer & Email Notification

Activity 1: Auto Classification Flow Design

- Created Flow: *Auto Classify School IT Tickets* triggered upon Record Creation when Category is empty.
- **Wi-Fi Rule:** If Short Description contains 'Wi-Fi' or 'Network' → Set Category = Network, Subcategory = Wi-Fi.
- **Hardware Rule:** If Short Description contains 'Projector' → Set Category = Hardware, Subcategory = Projector.
- **Access Rule:** If Short Description contains 'Password' → Set Category = Access, Subcategory = Forgot Password.
- **Performance Rule:** If Short Description contains 'Slow' → Set Category = Performance, Subcategory = Slow Computer.
- **Email Action:** Automatically sends confirmation email ('Your Request for the issue has been submitted.') to Caller email.

5. Milestone 4: Testing, Validation & Security

Test Scenarios & Validation:

- **Test Scenario 1 (Network):** Entered Short Description 'WiFi not working in library'. Verified Category auto-filled as *Network* and Subcategory as *Wi-Fi* with confirmation email sent.
- **Test Scenario 2 (Email Logs):** Verified email logs under System Logs → Emails and checked preview mail.
- **Test Scenario 3 (Hardware):** Entered Short Description 'Projector not turning on'. Verified Category auto-filled as *Hardware* and Subcategory as *Projector*.

6. Milestone 5: Deployment & Conclusion

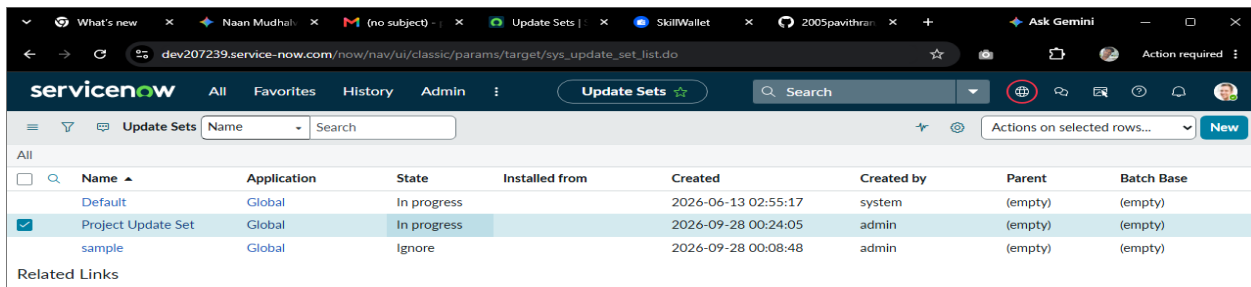
Activity 1: Finalization & XML Export

- Navigated to Local Update Sets, changed state from *In progress* to **Complete**, saved, and exported to XML for deployment.

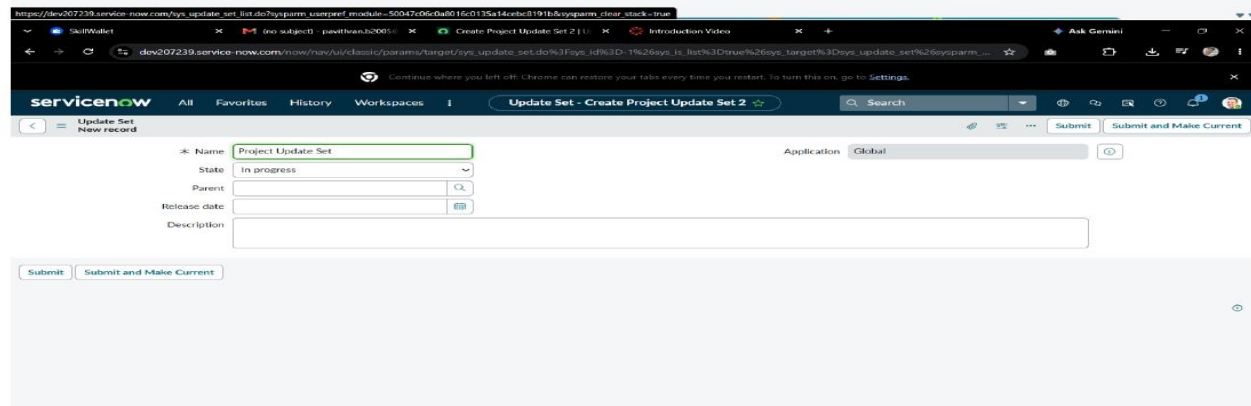
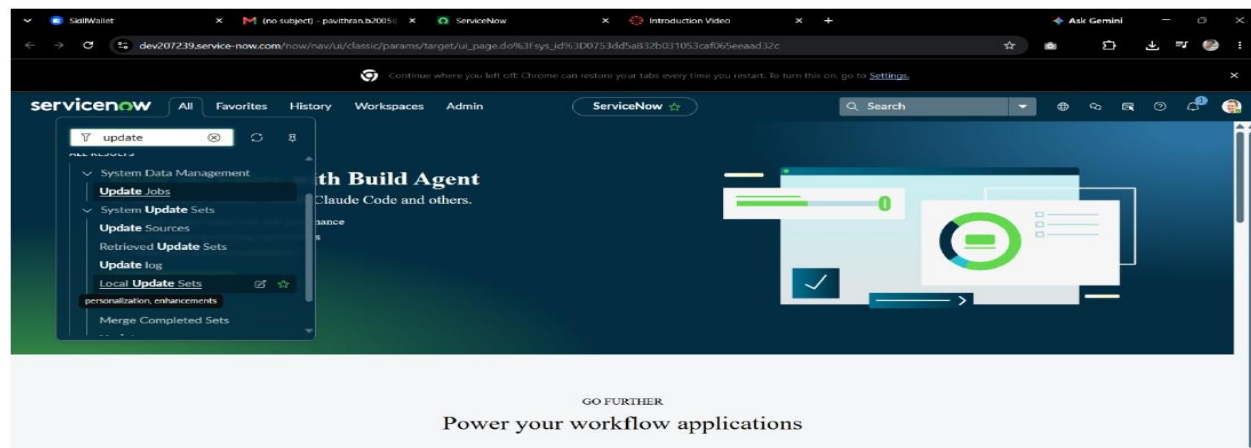
Conclusion: The project successfully automates school IT helpdesk ticketing, eliminating manual triage and ensuring rapid, accurate request routing.

Screenshots

Output:



	Name	Application	State	Installed from	Created	Created by	Parent	Batch Base
☐	Default	Global	In progress		2026-06-13 02:55:17	system	(empty)	(empty)
☒	Project Update Set	Global	In progress		2026-09-28 00:24:05	admin	(empty)	(empty)
☐	sample	Global	Ignore		2026-09-28 00:08:48	admin	(empty)	(empty)



Update Set - Create Project Update Set 2

Name: Project Update Set

State: In progress

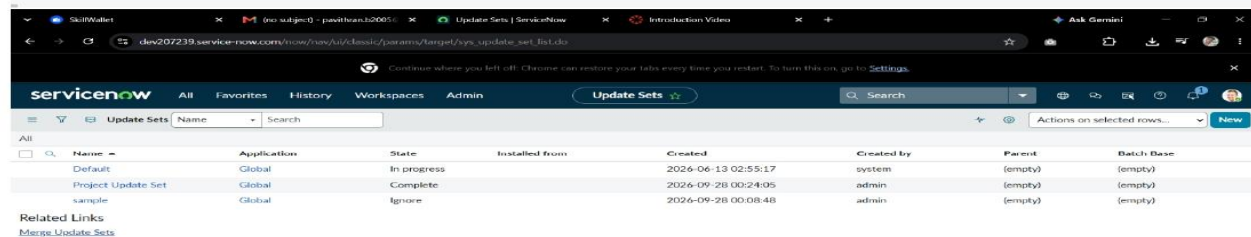
Parent:

Release date:

Description:

Application: Global

Submit Submit and Make Current



	Name	Application	State	Installed from	Created	Created by	Parent	Batch Base
☐	Default	Global	In progress		2026-06-13 02:55:17	system	(empty)	(empty)
☐	Project Update Set	Global	Complete		2026-09-28 00:24:05	admin	(empty)	(empty)
☐	sample	Global	Ignore		2026-09-28 00:08:48	admin	(empty)	(empty)

Table Incident WorkFlow

* Label Incident WorkFlow

* Name u_incident_workflow

Application Global

Remote Table

Columns Controls Application Access

Table Columns Type Search

Dictionary Entries

Column label	Type	Reference	Max length	Default value	Display
Created	Date/Time	(empty)	40		false
Updated	Date/Time	(empty)	40		false
Updates	Integer	(empty)	40		false
Assigned group	Reference	Group	32		false
Caller	Reference	User	32		false
Assigned to	Reference	User	32		false
State	String	(empty)	40	new	false
Updated by	String	(empty)	40		false

Form Design

Incident WorkFlow [u_incident_workflow]

Fields

Field Types

Filter

Fields

Created

Created by

Updated

Updated by

Updates

Formatters

Activities (filtered)

Contextual Search Results

Ratings

Properties

Dependent

Choices

Choice type Dropdown with none

Network

Hardware

Access

Performance

State

Assigned group

Assigned to

Create INC0001022 | Incident WorkFlow | ServiceNow - Google Chrome

dev207239.service-now.com/u_incident_workflow.do?&sysparm_clear_stack=true&sysparm_reference=u_incident_workflow&...

Incident WorkFlow
New record

Submit

Number: INC0001022

State: New

Caller:

Assigned group:

Category: Network

Assigned to:

Subcategory: -- None --

Short Description: -- None --

Description:

Submit

Incident WorkFlow New record

Number

Call

Category

Subcategory

Short Description

Description

Submit

Create INC0001022 | Incident WorkFlow | ServiceNow - Google Chrome

dev207239.service-now.com/u_incident_workflow.do?&sysparm_clear_stack=true&sysparm_reference=u_incident_workflow&...

Incident WorkFlow
New record

Submit

Number: INC0001022

State: New

Caller:

Assigned group:

Category: Network

Assigned to:

Subcategory: -- None --

Short Description: Network

Description: Hardware

Access

Performance

Submit

Incident WorkFlow New record

Number

Call

Category

Subcategory

Short Description

Description

Submit

Number: INC0001031
Caller: Chad Miklas
Category: Network
Subcategory: Wi-Fi
Short Description: My college has issue with Network
Description:
State: New
Assigned group:
Assigned to:

Submit

Preview Email

Hi greetings from AURCC...!!!
We are viewed your request and solved by short time period.,
Thankyou,
AURCC.

Ticket confirmation message

Ref:MSG0000288_txvnNGLdTRSijcIIdHnS

Close

Update Delete
Related Links
Preview Email

Email Log Email Attachments

Created Search

Email Log

Created



No records to display

SkillWallet

(no subject) - pavithran.b2005@

Create INC0001032 | Incident V

Introduction Video

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Ask Gemini

dev207239.service-now.com/now/nav/ui/classic/params/target/u_incident_workflow.do%3Fsys_id%3D-1

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Incident Workflow - Create INC0001032 ☆

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Incident Workflow

New record

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Submit

Number

INC0001032

State

New

⌵

Caller

Audra Cantu

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Assigned group

🔍

Category

Hardware

⌵

Assigned to

🔍

Subcategory

Projector

⌵

Short Description

Project is not turning on

Description

Submit

🔗

Conclusion

- **The Auto Ticket Classification using Flow Designer project successfully delivers a complete, end-to-end automation solution for a school IT helpdesk. The primary objective of reducing manual effort in ticket categorization and improving response efficiency has been fully achieved.**
- **Through structured requirement analysis, careful backend configuration, and no-code automation using Flow Designer, the system now automatically classifies IT tickets based on the short description provided by students. The implementation of dependent choice fields ensures data accuracy, while automated email notifications improve communication by instantly acknowledging ticket creation to the caller.**
- **The solution was developed using a phased approach, ensuring clarity, stability, and ease of maintenance. Comprehensive testing and security validation confirmed that the system functions reliably, maintains data integrity, and enforces appropriate access controls.**
- **Overall, this project demonstrates how ServiceNow can be effectively used to build scalable, maintainable, and user-friendly automation solutions without relying on complex scripting or machine learning. The design is well-suited for small to medium environments such as educational institutions and can be easily extended in the future to include advanced features like assignment automation, SLA tracking, or predictive intelligence.**